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Virtual-Beacon Depth Refocusing for Improved Received-Power Uniformity in Near-Field Time-Reversal Wireless Power Transfer

Young-Seok Lee, *Graduate Student Member, IEEE*, Sihyun Kim, *Undergraduate Student Member, IEEE*, Taeyeong Yoon, *Graduate Student Member, IEEE*, Sangwook Nam, *Life Senior Member, IEEE*, and Jungsuek Oh, *Senior Member, IEEE*

***Abstract*—This letter proposes a virtual-beacon-based depth-refocusing technique that improves the received-power uniformity of near-field time-reversal (TR) wireless power transfer (WPT). For an array receiver (RXA), conventional TR focuses the field back onto the beacon, and the resulting high non-uniformity, which especially worsens for larger arrays, keeps the array rectifiers from operating efficiently and limits the received DC power. Inspired by an optical lens, the proposed method adjusts a single coefficient β to refocus the beam toward a 'virtual beacon' in depth, spreading the incident power more uniformly over the RXA. Since the refocusing acts directly on the measured single-beacon phase surface, it requires no RXA localization and retains the real-time, low-complexity phase processing of TR. In a 5.64 GHz scenario, simulations and measurements confirm a substantial uniformity improvement, raising the received DC power by up to 158%. A feasible feedback-based closed-loop search for real-time implementation is further proposed.**

***Index Terms*—Beam-focusing, Near field, Time reversal.**

I. Introduction

RADIO Frequency-based Wireless Power Transfer (WPT) has been studied since it can deliver power over long distances and direct energy toward the receiver [1], [2]. One of the efficient beamfocusing/forming algorithms used in WPT is time-reversal (TR), effective for various applications due to its low computational cost and real-time operation [3] while still achieving good performance [4]. Even in a single-element receiver, it provides optimal power transfer efficiency (PTE) [5], [6]. However, when the receiver is an array (RXA), the RF power is focused onto the beacon, resulting in non-uniformity in the received RF power pattern over the RXA [7]. This causes a critical issue in the rectifier RF-to-DC conversion, because practical rectifiers exhibit maximum efficiency only within a certain input-power range; therefore, due to the non-uniformity, the received power of all array elements cannot fall within the high-efficiency range of rectifiers simultaneously, leading to inefficient utilization of the receiving aperture.

To address this problem, rectifier-level approaches have been proposed. Studies have been conducted to expand the rectifier input dynamic range [8], [9], [10]. Although these approaches mitigate the efficiency degradation, they remain limited under highly non-uniform received power.

Another approach is to control the transmitter array (TXA) excitation directly to obtain more uniform RF power distribution at the RXA. Flat-top beamforming has been studied in the context of far-field WPT using the Woodward–Lawson method [11], [12], [13], or via beam synthesis using S-parameters [14]. However, these methods generally require prior knowledge of the RXA geometry or the TXA–RXA channel, which limits their applicability to real-time moving-RXA scenarios.

Inspired by the optical principle that the focal distance of a convex lens can be changed by varying only its radius of curvature R (assuming a constant refractive index), we propose a method that *intentionally refocuses the beam toward an ideal location of a 'virtual beacon' in depth* by controlling only a single parameter β. In this way, the received RF power across the RXA is deliberately defocused into a more uniform distribution, letting the array rectifiers operate efficiently and thus raising the received DC power. Since the refocusing is applied directly to the measured single-beacon phase surface, no explicit RXA localization is required. Moreover, the proposed method adopts the simplest hardware structure of a TR-based system, with a single beacon at the RXA and the TXA element powers uniform and only the phases controlled. Due to its low computational complexity, feasiblity for real-time implement application is also proposed.

II. Intentional Refocusing Technique

Fig. 1 compares the single-beacon TR-WPT system under conventional and proposed operation. In both cases, a beacon at the center of the RXA transmits a pilot signal toward the TXA, which extracts the channel from the received pilot. As shown in Fig. 1(a), conventional TR then immediately conjugates the measured phase, resulting in focusing onto the physical beacon, whereas the proposed scheme in Fig. 1(b) processes the phase before conjugation to refocus the beam

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Young-Seok Lee, Taeyeong Yoon, Sangwook Nam, and Jungsuek Oh are with the Institute of New Media and Communication (INMC) and the Department of Electrical and Computer Engineering, Seoul National University, Seoul 08826, Republic of Korea (e-mail: ryanlee@snu.ac.kr; taeyeong.yoon@snu.ac.kr; snam@snu.ac.kr; jungsuek@snu.ac.kr).

Sihyun Kim is with the School of Electrical and Electronic Engineering, Yonsei University, Seoul 03722, Republic of Korea (e-mail: sihyunkim02@yonsei.ac.kr).

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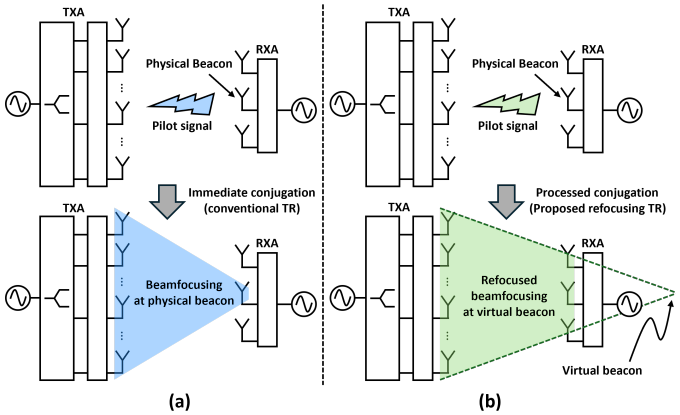


Fig. 1. (a) Conventional and (b) Proposed depth-refocusing TR-WPT system.

in depth onto a virtual beacon, spreading the incident power more uniformly over the aperture.

A. Virtual-Beacon Refocusing

Let $R(x,y)$ denote the distance from the physical beacon (PB, located at RXA center) to an arbitrary point $r=(x,y,0)$ on the $M\times M$ sized TXA plane as Fig. 2. The absolute physical phase surface of the pilot signal observed at r is

$$\phi(x,y)=k\{R(x,y)-F_{\text{phy}}\}+C \tag{1}$$

where $k=2\pi/\lambda$, C is the common phase offset, and F_{phy} is the distance from the TXA center to PB. Set PB location $\mathbf{r}_{\text{phy}}=F_{\text{phy}}\hat{\mathbf{u}}=(F_{\text{phy}}u_x,F_{\text{phy}}u_y,F_{\text{phy}}u_z)$, then:

$$R(x,y)=\left[F_{\text{phy}}^2-2F_{\text{phy}}(u_xx+u_yy)+x^2+y^2\right]^{1/2}. \tag{2}$$

Letting $s=u_xx+u_yy$ and $q=x^2+y^2$, (2) can be written as

$$R(x,y)=F_{\text{phy}}\left(1-\frac{2s}{F_{\text{phy}}}+\frac{q}{F_{\text{phy}}^2}\right)^{1/2}. \tag{3}$$

By applying a fourth-order Taylor approximation[1], the path difference can be expressed as

$$R(x,y)-F_{\text{phy}}\simeq -s+\frac{q-s^2}{2F_{\text{phy}}}+\frac{s(q-s^2)}{2F_{\text{phy}}^2}+\frac{-q^2+6s^2q-5s^4}{8F_{\text{phy}}^3}. \tag{4}$$

The first-order term, independent of F_{phy}, means the direction of the beacon, whereas the higher order terms, proportional to $1/F_{\text{phy}}$, $1/F_{\text{phy}}^2$, and $1/F_{\text{phy}}^3$, represent the curvature terms. For simplicity, the equation can be expressed using a relative phase surface $\psi\triangleq\phi-\phi(0,0)=k\{R-F_{\text{phy}}\}$:

$$\psi_{\text{phy}}^{(4)}(x,y)=H_1(x,y)+H_2(x,y)+H_3(x,y)+H_4(x,y) \tag{5}$$

where $H_i(x,y)$ denotes the i-th order term of (4) multiplied by k. Now *refocusing coefficient* 'β' is defined as

$$\beta=\frac{F_{\text{phy}}}{F_{\text{vir}}}\begin{cases}\beta=1 & : \text{ Conventional TR}\\ \beta>1 & : \text{ Closer Refocus}\\ 0<\beta<1 & : \text{ Farther Refocus}\end{cases} \tag{6}$$

[1]The expansion order can be selected for TXA aperture, range, and steering angle; larger apertures require higher orders due to increased phase variation, whereas fourth order suffices for the operating scenarios considered here.

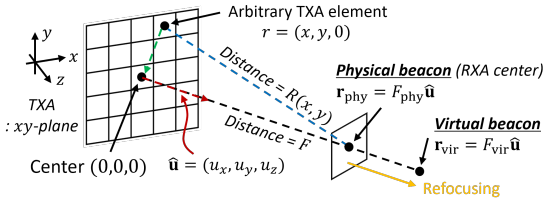


Fig. 2. Geometry and its notations for the proposed TR-WPT system.

The modified phase surface is obtained by preserving the first-order direction term and scaling only the higher-order near-field curvature terms as:

$$\begin{aligned}\psi_{\text{vir}}^{(4)}(x,y;\beta)=H_1(x,y)+\beta H_2(x,y)\\ +\beta^2H_3(x,y)+\beta^3H_4(x,y)\end{aligned} \tag{7}$$

each $H_i(x,y)$ is a homogeneous polynomial and satisfies:

$$H_i(\beta x,\beta y)=\beta^iH_i(x,y). \tag{8}$$

Therefore, the proposed virtual phase surface is obtained as

$$\frac{1}{\beta}\psi_{\text{phy}}^{(4)}(\beta x,\beta y)=\psi_{\text{vir}}^{(4)}(x,y;\beta). \tag{9}$$

For $0<\beta<1$, this transformation admits an intuitive optical-lens interpretation. The measured phase surface $\psi_{\text{phy}}^{(4)}$ acts as a converging lens that focuses the transmitted field onto the physical beacon. Forming $\beta^{-1}\psi_{\text{phy}}^{(4)}(\beta x,\beta y)$ keeps only the gently curved central portion of this lens and stretches it across the entire TXA aperture, yielding an effective lens with a larger R. Since a flatter lens has a longer focal length, the focus is pushed behind the physical beacon, so the field spreads the power more uniformly across the aperture.

B. Coefficient Calculation From Phase Gradients

Transforming $\psi_{\text{phy}}^{(4)}(\beta x,\beta y)$ from $\psi_{\text{phy}}^{(4)}(x,y)$ is required. Full expansion of the fourth-order phase surface $\psi_{\text{phy}}^{(4)}(x,y)$ from equation (4) and (5) can be written as:

$$\psi_{\text{phy}}^{(4)}(x,y)=c_{10}x+c_{01}y+c_{20}x^2+\cdots+c_{13}xy^3+c_{04}y^4 \tag{10}$$

where the coefficient vector is defined as

$$\mathbf{c}=[c_{10},c_{01},c_{20},\ldots,c_{22},c_{13},c_{04}]^T \tag{11}$$

Since using the wrapped phases requires complex phase unwrapping, c_{ij} are instead determined from the phase differences between adjacent TXA elements. Let $\psi_{i,j}$ denote the pilot phase measured at the (i,j)-th TXA element. Then, the phase differences between adjacent elements are calculated as

$$g_{i,j}^{(x)}=\angle(e^{j(\psi_{i+1,j}-\psi_{i,j})}),\ g_{i,j}^{(y)}=\angle(e^{j(\psi_{i,j+1}-\psi_{i,j})}) \tag{12}$$

these differences can be approximated as the phase gradients at the midpoints between adjacent elements. By differentiating (10), the gradient can be approximated at the midpoint as:

$$\frac{g_{i,j}^{(x)}}{d}\simeq\left[\frac{\partial\hat{\psi}_{\text{phy}}^{(4)}(x,y)}{\partial x}\right]_{x\leftarrow\bar{x}},\ \frac{g_{i,j}^{(y)}}{d}\simeq\left[\frac{\partial\hat{\psi}_{\text{phy}}^{(4)}(x,y)}{\partial y}\right]_{y\leftarrow\bar{y}} \tag{13}$$

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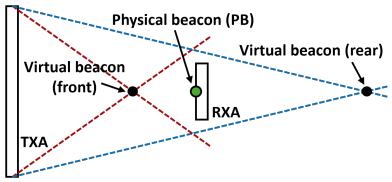


Fig. 3. Geometric illustration of front and rear virtual beacons

where $\bar{x}$ and $\bar{y}$ denote the x and y midpoints, respectively. The x, y-direction phase difference vectors are,

$$\mathbf{a}_{i,j}^{(x)} = [1, 0, 2\bar{x}_i, y_j, 0, 3\bar{x}_i^2, 2\bar{x}_i y_j, y_j^2, 0, 4\bar{x}_i^3, 3\bar{x}_i^2 y_j, 2\bar{x}_i y_j^2, y_j^3, 0] \in \mathbb{R}^{1\times 14} \tag{14}$$

$$\mathbf{a}_{i,j}^{(y)} = [0, 1, 0, x_i, 2\bar{y}_j, 0, x_i^2, 2x_i\bar{y}_j, 3\bar{y}_j^2, 0, x_i^3, 2x_i^2\bar{y}_j, 3x_i\bar{y}_j^2, 4\bar{y}_j^3] \in \mathbb{R}^{1\times 14}, \tag{15}$$

respectively, and stacking all gradient observations over the TXA aperture, an overdetermined linear system is obtained as

$$\mathbf{A}\cdot\mathbf{c} = \mathbf{b} \tag{16}$$

where $\mathbf{A} \in \mathbb{R}^{2M(M-1)\times 14}$ and $\mathbf{b} \in \mathbb{R}^{2M(M-1)\times 1}$ are

$$\mathbf{A} = \begin{bmatrix} (\mathbf{a}_1^{(x)})^T & (\mathbf{a}_2^{(x)})^T & \cdots & (\mathbf{a}_1^{(y)})^T & \cdots \end{bmatrix}^T$$
$$\mathbf{b} = \begin{bmatrix} g_1^{(x)}/d & g_2^{(x)}/d & \cdots & g_1^{(y)}/d & \cdots \end{bmatrix}^T \tag{17}$$

The $\mathbf{c}$ is then estimated by the least-squares (LS) solution:

$$\hat{\mathbf{c}} = \arg\min_{\mathbf{c}} \|\mathbf{A}\mathbf{c} - \mathbf{b}\|_2^2 = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b} \tag{18}$$

where $\hat{\mathbf{c}}$ is the estimated coefficient $\mathbf{c}$. The polynomial terms $H_i(x, y)$ then construct $\psi_{\text{vir}}^{(4)}(x, y; \beta)$ in (7); its conjugate $\sqrt{P_{\text{tx}}}e^{-j\psi_{\text{vir}}^{(4)}(x,y;\beta)}$ is applied as the TXA excitation, identical to conventional channel-conjugate TR ($\beta = 1$). Direct fitting avoids localization error, while the higher-order terms absorb non-idealities that a point-source estimate would miss.

III. Simulation Results

A. Setup and Design-Space Analysis

The proposed method is evaluated with a free-space simulation implemented in Python and executed on an AMD Ryzen 7 5800X, with the TXA–RXA channel modeled by the Friis equation [15]. The operating frequency is 5.64 GHz, the element spacing of both arrays is 0.6λ, and the TXA and RXA sizes are 16×16 and 5×5, matching the hardware introduced in Section IV. A single beacon is at the center of the RXA, and the arrays are aligned at distance F along the boresight.

Each RXA element feeds a half-wave Schottky rectifier whose measured RF-to-DC characteristic (peak PCE 45.4% at 23 dBm, characterized over 0–35 dBm) is fitted by a piecewise-linear (PWL) function [16], [17], [18], [19] and used in the simulation. Assuming ideal combining, the total received DC power $P_{\text{RXA}}^{\text{DC}}$ is the sum of each rectifier's output. Two efficiency metrics from [20] are adopted,

$$\eta_{\text{TCE}} = \frac{P_{\text{RXA}}^{\text{DC}}}{P_{\text{TXA}}^{\text{rad}}}, \quad \eta_{\text{RUE}} = \frac{P_{\text{RXA}}^{\text{DC}}}{N^2\cdot P_{\text{rect}}^{\text{peak}}}, \tag{19}$$

TABLE I
Simulated Received DC Power and Efficiencies.

Scenario	β	$P_{\text{RXA}}^{\text{DC}}$ [W]	RUE [%]	TCE [%]	Enhancement
On-axis (Conv.)	1.00	3.70	25.9	4.6	–
On-axis (Refoc.)	0.50	12.07	84.4	14.9	+226%
30° (Conv.)	1.00	6.36	44.4	7.9	–
30° (Refoc.)	0.55	9.21	64.4	11.4	+45%

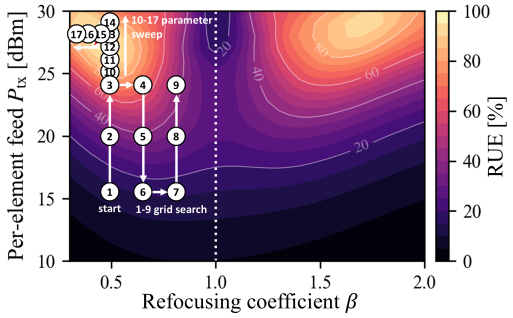


Fig. 4. RUE versus β and P_{tx}. The dotted line marks conventional TR ($\beta = 1$); the star marks the maximum ($\eta_{\text{RUE}} = 98.8\%$ at 28 dBm, $\beta = 0.45$).

where $P_{\text{TXA}}^{\text{rad}}$ is the total radiated RF power at the TXA, $P_{\text{rect}}^{\text{peak}} = 0.572$ W is the rectifier peak DC output, and N^2 is the number of RXA elements. The transmission–conversion efficiency (TCE) is the overall RF-to-DC efficiency, while the receiver utilization efficiency (RUE) measures how fully the RXA aperture is used. Since term $N^2\cdot P_{\text{rect}}^{\text{peak}}$ is fixed for a given rectenna, RUE is proportional to $P_{\text{RXA}}^{\text{DC}}$; RUE and the received DC power are thus equivalent.

Since β sets the beam-depth, it can be refocused in front of ($\beta > 1$) or behind ($\beta < 1$) the RXA, as shown in Fig. 3. This can also provide a selectable degree of freedom, selectively by scenarios. Fig. 4 maps the RUE over β and the TXA per-element feed P_{tx}. Due to its strong focusing, conventional TR ($\beta = 1$) drives most elements far from the rectifier's PCE peak, therefore the distribution is highly non-uniform (low RUE) thus $P_{\text{RXA}}^{\text{DC}}$ drops sharply. Refocusing in either direction spreads the beam; the rear branch ($\beta < 1$) gives the higher peak, reaching $\eta_{\text{RUE}} = 98.8\%$ at $\beta = 0.45$.

B. Baseline Scenario Setup and Interpretation

The baseline scenario places the RXA on-axis at a distance of 0.5 m, with a beam-steered case at an azimuth angle of 30° and 0.7 m. The uniform feed power is set to $P_{\text{tx}} = 25$ dBm matching the experiments, and the optima β were 0.50 and 0.55, respectively. Fig. 5 shows the xz-plane normalized E-field (Top-view), contrasting conventional TR with refocusing: the conventional beam focuses tightly on the RXA, whereas refocusing pushes the focus behind it, spreading the illumination across the aperture. The same behavior holds under beam-steered, confirming the generality of this method.

Table I quantifies the resulting gains: refocusing raises RUE from 25.9% to 84.4% on-axis (44.4% to 64.4% steered); equivalently, the received DC power rises by 226% (45%), from 3.70 to 12.07 W on-axis, at computation time (CT) = 22 μs per refocusing. RUE and TCE trade off through P_{tx}: a lower feed would raise TCE but underuse the aperture, so

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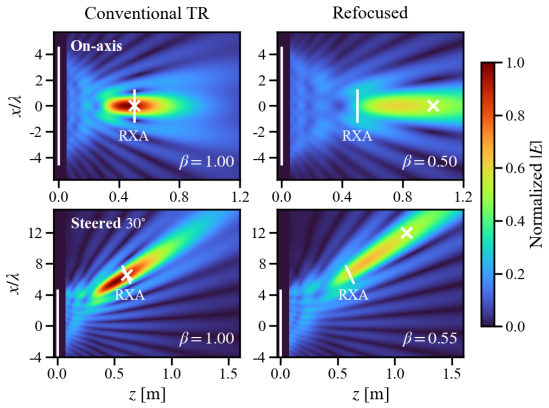


Fig. 5. Normalized $|E|$ on the xz-plane for conventional TR (left) and refocusing (right), under on-axis (top) and 30°-steered (bottom) operation.

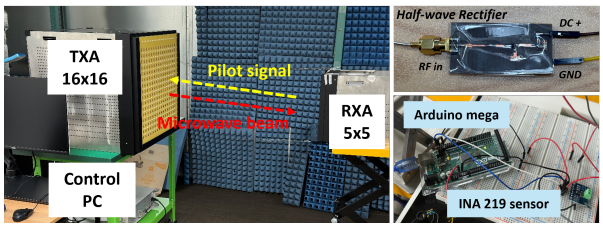


Fig. 6. Measurement setup for the proposed refocusing technique.

TABLE II
RECEIVED DC POWER [W]: SIMULATION VERSUS MEASUREMENT.

	Conv.		Refoc.		Enhancement	
Scenario	Sim.	Meas.	Sim.	Meas.	Sim.	Meas.
On-axis	3.70	4.21	12.07	10.86	+226%	+158%
30°-steered	6.36	6.72	9.21	8.24	+45%	+23%

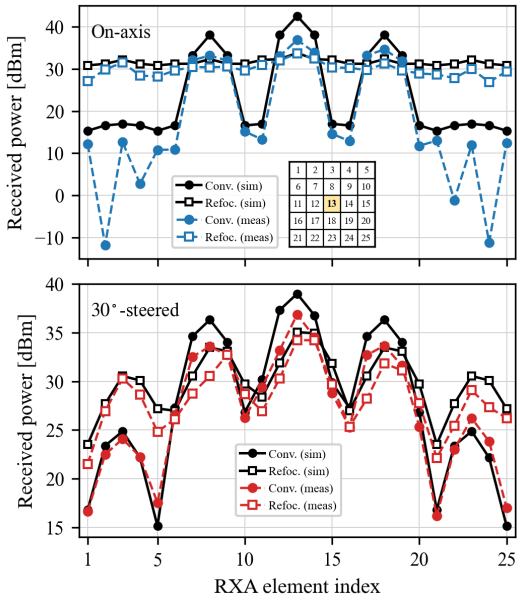


Fig. 7. RXA received power per element (inset: front-view 5×5 RXA indexing; beacon: 13). Conventional TR (circles) and the proposed refocusing (squares); simulated (solid) and measured (dashed).

25 dBm prioritizes utilization. Such control of the received distribution is meaningful only in the near field; in far-field, the RXA is illuminated uniformly thus RUE is not usable.

C. Proposed Closed-Loop Search for Real-Time Operation

Our present hardware does not support real-time feedback, so the following closed-loop search is proposed rather than demonstrated. With unknown RUE map in real-world, a sparse (β, P_{tx}) grid is tested from the fed-back DC power for initial operating point. From its best point, P_{tx} is swept at fixed β, followed by a β sweep at the updated P_{tx}. Fig. 4 shows the example trial of this feedback; the count and sweep trajectory are illustrative, not fixed. With the rapid phase-refocusing computation, this search supports the feasibility of real-time implementation using suitable system hardware.

IV. EXPERIMENTAL RESULTS

Fig. 6 shows the measurement setup, whose hardware follows the same specifications as the simulation (16×16 TXA, 5×5 RXA, $d = 0.6\lambda$, 5.64 GHz, and uniform 25 dBm). The TXA is fully designed and calibrated [21], [6]. A single beacon, set at the center element of the RXA, transmits a pilot signal that is detected at the TXA from which the refocused excitation is computed and applied through digital controls [22]. The RF power received at each RXA port is then measured with a spectrum analyzer for the two scenarios matching the simulation: on-axis ($F = 0.5$ m, $\beta = 0.50$) and 30°-steered ($F = 0.7$ m, $\beta = 0.55$).

Fig. 7 compares the measured element received RF power for conventional TR and the proposed refocusing. Conventional TR yields a highly non-uniform distribution with deep nulls, whereas refocusing markedly flattens the received RF power, reducing the measured dynamic range from 48.6 to 6.8 and from 20.7 to 12.7 dB for the on-axis and 30°-steered cases, respectively. The measured refocused results (dashed lines) closely follow the simulation (solid lines), validating the technique in practice.

The DC power is measured individually with a power sensor (INA 219) using arduino at each rectifier and summed assuming ideal DC combining. As presented in Table II, refocusing significantly improves the DC power, by 158% on-axis and 23% when steered. The measured enhancement is smaller than the simulated one; due to the hardware imperfections, power is scattered out of the focus, and the conventional TR peak is lowered from 42.5 dBm to 36.9 dBm. This influences the rectifier output and reduces the enhancement, but the measured results still confirm the effectiveness of the proposed method.

V. CONCLUSION

This letter proposed a depth-refocusing technique using a virtual beacon that improves the received-power uniformity of near-field TR-based WPT with a single coefficient β, while retaining the low-complexity nature of conventional TR. Simulations and measurements confirmed the improvement for both on-axis and beam-steered operation, raising the received DC power by up to 158%. Due to its low computational complexity, a feasible feedback-based closed-loop search for real-time implementation was also proposed.

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